

Thinking Machines, Ancient Minds

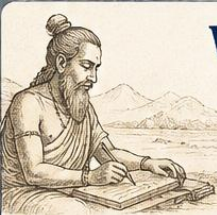
Toward an Indian Cognitive Architecture for Artificial Intelligence

Article 2 (Part I)

What Ancient India Thought the Mind Actually Is

Beyond the Myth of the Single Mind

"The mind is not a thing. It is an organization."



WHAT ANCIENT INDIA THOUGHT THE MIND ACTUALLY IS

Four Functional Components. One Integrated System.

“The mind is not a thing. It is an organization.”

मनस् (MANAS)

The Coordinator of Attention



- Receives input from the senses
- Selects, filters, and prioritizes
- Directs attention and coordinates
- Handles moment-to-moment mental activity

Key Role:
Gateway of cognition.
What gets attention, gets processed.

बुद्धि (BUDDHI)

The Faculty of Discernment and Decision



- Discriminates, evaluates, and judges
- Solves problems and reasons
- Decides and commits to action
- Seeks truth and determines what is right

Key Role:
Makes choices. Determines direction and meaning.

चित् (CHITTA)

The Field of Memory and Experience



- Stores experiences and impressions
- Provides long-term memory
- Shapes habits, preferences and emotional tendencies
- Supplies context for present cognition

Key Role:
The storehouse of experience.
Past shapes the present.

अहंकार (AHAMKARA)

The Organizer of Self and Agency

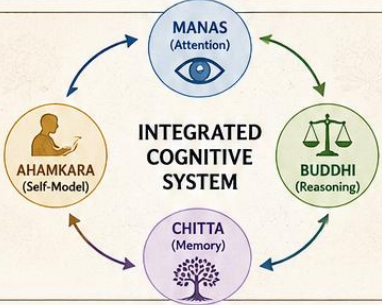


- Creates the sense of “I”
- Maintains identity and continuity
- Links experiences, goals and actions to the self
- Supports ownership and agency

Key Role:
Centers experience.
Answers: “I am the one who experiences.”

NOT FOUR ISLANDS, BUT ONE LIVING SYSTEM

These four functions are deeply interconnected. Attention influences memory. Memory guides reasoning. Reasoning shapes future memories. The self organizes goals across time. Together they create intelligent behavior.



INTEGRATED COGNITIVE SYSTEM

WHY THIS MATTERS FOR AI

Modern AI excels at computation but struggles with memory, reasoning, and self-modeling.

This ancient functional decomposition offers a timeless blueprint for cognitive architecture.

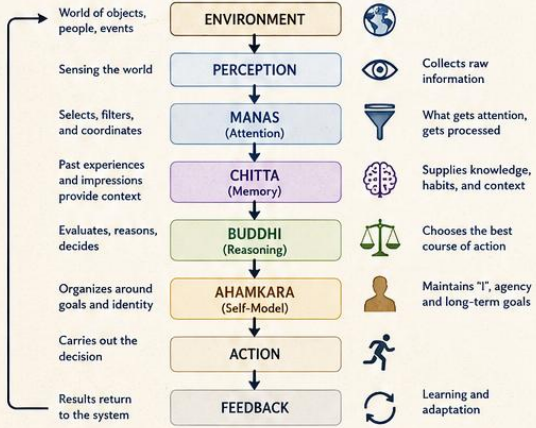
The future of AI may depend not only on scaling models, but on organizing cognition.

COMPARISON WITH MODERN COGNITIVE SCIENCE

Classical Concept	Functional Role	Modern Analogy
 मनस् (Manas)	Coordinates sensory input, selects and directs attention	Selective attention, attentional control, working memory (coordination)
 बुद्धि (Buddhi)	Judgement, discrimination, reasoning, decision-making	Executive control, reasoning, planning, cognitive flexibility
 चित् (Chitta)	Memory, impressions, habits, accumulated experience	Long-term memory, episodic memory, continual learning
 अहंकार (Ahankara)	Self-reference, identity, agency, ownership of actions	Self-model, agent identity, self-representation, autonomous agents

i These are functional analogies, not historical equivalences. They invite dialogue across traditions to inspire better architectures.

A CONCEPTUAL INDIAN COGNITIVE ARCHITECTURE (ICA)



World of objects, people, events → **ENVIRONMENT** → **PERCEPTION** → **MANAS (Attention)** → **CHITTA (Memory)** → **BUDDHI (Reasoning)** → **AHAMKARA (Self-Model)** → **ACTION** → **FEEDBACK** → back to ENVIRONMENT

Left side labels: Sensing the world, Selects, filters, and coordinates, Past experiences and impressions provide context, Evaluates, reasons, decides, Organizes around goals and identity, Carries out the decision, Results return to the system.

Right side labels: Collects raw information, What gets attention, gets processed, Supplies knowledge, habits, and context, Chooses the best course of action, Maintains “I”, agency and long-term goals, Learning and adaptation.



KEY TAKEAWAY

Classical Indian thinkers recognized that what we call “the mind” is an organized system of interacting functions. This insight—remarkably ancient yet strikingly modern—may help us design the next generation of intelligent machines.



QUESTION FOR THE NEXT ARTICLE

How should an intelligent system determine what is true? The Nyāya school answered this through a sophisticated theory of perception, inference, analogy, and testimony—ideas deeply relevant to modern AI and explainable reasoning.

We Speak of "The Mind" as Though It Were One Thing

Imagine standing in front of a busy airport.

Passengers arrive from every corner of the world. Security officers inspect luggage. Air-traffic controllers coordinate hundreds of aircraft. Maintenance engineers monitor equipment. Information screens update continuously. Restaurants serve travellers. Emergency teams remain on standby.

Despite this extraordinary complexity, we casually refer to the entire enterprise as "the airport."

No engineer would design an airport as a single undifferentiated building in which every activity occurred simultaneously.

Its efficiency depends upon specialization.

Different components perform different functions.

More importantly, they communicate continuously.

Now consider how we usually speak about the human mind.

We say:

"The mind remembers."

"The mind decides."

"The mind imagines."

"The mind pays attention."

"The mind reasons."

The language is convenient.

It is also misleading.

Memory does not appear to be identical to attention.

Attention is not identical to reasoning.

Reasoning differs from decision-making.

Decision-making differs from the experience of self.

Modern cognitive science increasingly views intelligence as the coordinated activity of multiple interacting systems rather than the operation of a single homogeneous processor.

Surprisingly, this way of thinking is not entirely new.

More than two thousand years ago, several schools of Indian philosophy proposed that what we casually call "the mind" is better understood as a collection of distinct yet interacting cognitive functions.

They disagreed on many philosophical questions.

Yet they shared one remarkably modern intuition:

Cognition is organized.

Not because they possessed neuroscience.

Not because they anticipated artificial intelligence.

But because careful observation suggested that different mental activities behave differently.

This insight may have profound implications for the future design of intelligent machines.

Why This Question Matters for Artificial Intelligence

One of the central assumptions underlying many modern AI systems is deceptively simple.

A sufficiently large computational system, trained on enough data, may develop increasingly sophisticated intelligence.

The extraordinary success of large language models demonstrates how powerful this assumption can be.

Yet engineers are gradually encountering a different class of questions.

How should memory interact with reasoning?

How should attention select relevant information?

How should long-term goals influence immediate actions?

How should an intelligent agent distinguish between important and irrelevant experiences?

How should it maintain a coherent sense of identity while continuously learning?

These questions are not primarily about computation.

They are questions about organization.

They concern architecture rather than scale.

As argued in the previous article, biology suggests that intelligence emerges not merely from increasing computational capacity but from organizing specialized cognitive functions into coherent systems.

If this observation is correct, then one of the most valuable sources of inspiration may come from traditions that devoted sustained attention to analysing the functional organization of cognition.

This is where classical Indian thought becomes unexpectedly relevant.

A Different Way of Asking the Question

Modern neuroscience often begins with the brain.

It asks:

Which neural circuits perform particular cognitive functions?

Classical Indian philosophy began somewhere else.

It asked:

What kinds of mental activities appear to exist?

This difference is important.

The first investigates implementation.

The second investigates function.

Engineers routinely make the same distinction.

An aircraft designer first identifies the necessary functions:

- generate lift,
- provide propulsion,
- maintain stability,
- navigate,
- communicate,
- land safely.

Only afterwards does the engineer determine how those functions will be implemented.

Similarly, cognitive architecture asks:

What functions are necessary for intelligence?

Only later do neuroscientists or AI engineers determine how those functions might be realized.

Viewed in this light, classical Indian analyses of cognition resemble functional decompositions rather than neurological theories.

Their value lies not in anatomical accuracy but in conceptual organization.

Pull Quote

"Perhaps the most important question is not where intelligence occurs, but how its functions are organized."

Four Functional Components

Among several Indian philosophical traditions, particularly Sāṃkhya, Yoga, and Vedānta, one repeatedly encounters a remarkably elegant decomposition of cognition.

Instead of speaking about "the mind" as a single faculty, these traditions distinguish four closely interacting functional components:

- **Manas**
- **Buddhi**
- **Chitta**
- **Ahamkara**

These Sanskrit terms have often been translated simply as "mind," "intellect," "memory," and "ego."

Unfortunately, these translations can create misunderstandings.

Each concept represents a functional role rather than a discrete physical structure.

They should therefore be interpreted cautiously.

The purpose of this article is not to argue that these concepts correspond directly to specific brain regions or modern psychological constructs.

Nor is it to suggest that ancient philosophers anticipated cognitive neuroscience.

Instead, we shall ask a different question.

Can these functional distinctions inspire future cognitive architectures for artificial intelligence?

Manas: The Manager of Attention

Imagine reading a scientific paper in a crowded airport lounge.

Announcements echo through loudspeakers.

People walk past.

Conversations occur nearby.

Your phone vibrates.

An aircraft departs outside the window.

Yet somehow you remain focused on a single paragraph.

This ability appears so ordinary that we rarely notice its complexity.

Your nervous system receives vastly more information than reaches conscious awareness.

Some mechanism continuously selects.

It prioritizes.

It filters.

It shifts.

It redirects.

Without such selection, intelligent behaviour would become impossible.

Classical Indian philosophy assigned much of this coordinating role to **Manas**.

Although often translated as "mind," Manas functions more accurately as a coordinator of incoming information.

It mediates between the senses and higher cognitive processes.

It determines what receives attention.

It helps coordinate competing streams of sensory input.

In modern terminology, Manas bears intriguing functional similarities to mechanisms of selective attention and aspects of working memory control.

The comparison is not exact.

But it is conceptually suggestive.

Today's transformer architectures have demonstrated the extraordinary importance of attention mechanisms.

Perhaps this should not surprise us.

Long before machine learning, philosophers recognized that intelligence depends not only upon acquiring information but upon selecting what deserves further processing.

Attention is not merely another cognitive process.

It is the gateway through which all subsequent cognition must pass.

Buddhi: Beyond Calculation

Suppose two researchers examine exactly the same experimental results.

One simply reports the observations.

The other recognizes a previously unnoticed pattern that leads to an entirely new scientific theory.

Both possess information.

Only one demonstrates judgement.

This distinction lies at the heart of **Buddhi**.

In classical Indian philosophy, Buddhi represents discrimination, discernment, evaluation, and decision-making.

It is the faculty that weighs alternatives, resolves uncertainty, and determines appropriate action.

Importantly, Buddhi is not simply intelligence measured as computational speed.

It concerns the quality of reasoning.

The ability to distinguish.

To evaluate.

To choose.

Modern cognitive science might compare some aspects of Buddhi with executive control, cognitive flexibility, and higher-order reasoning.

Again, the comparison is functional rather than historical.

Yet it highlights an increasingly important question for AI.

Can future intelligent systems merely calculate?

Or must they also learn to judge?

That distinction may ultimately separate increasingly capable computational systems from genuinely intelligent cognitive architectures.

End of Part I

Coming in Part II

The remaining components—**Chitta** and **Ahamkara**—introduce two ideas that remain central to contemporary AI research: memory and the self-model. We will explore how classical Indian theories of memory and identity compare with modern concepts such as long-term memory, continual learning, self-representation, and autonomous agents. Finally, we will ask whether these four functional components together constitute one of history's earliest conceptual cognitive architectures and what lessons they might offer for the design of future AI systems.

Thinking Machines, Ancient Minds

Toward an Indian Cognitive Architecture for Artificial Intelligence

Article 2 (Part II)

What Ancient India Thought the Mind Actually Is

Chitta: Memory Beyond Storage

Imagine meeting a former classmate after twenty years.

The face appears familiar, yet the name refuses to emerge.

A few moments later, seemingly without effort, an entire cascade of memories unfolds.

You remember the classroom.

The conversations.

Shared projects.

The teacher's voice.

Even emotions long forgotten return with surprising clarity.

Memory is clearly doing far more than storing information.

It organizes experience.

It shapes expectations.

It influences decisions before we consciously realize it.

Modern neuroscience increasingly recognizes that memory is not a single repository but a dynamic system involving encoding, consolidation, retrieval, reconstruction, and continual updating.

Every new experience is interpreted through previous experience.

Artificial Intelligence has encountered remarkably similar challenges.

Large language models possess enormous quantities of learned statistical knowledge, yet they often struggle with continual learning, long-term personalization, episodic memory, and integrating new experiences without disrupting previous knowledge.

Memory remains one of the great unsolved problems in AI.

Classical Indian philosophy devoted considerable attention to this problem.

The concept of **Chitta** is often translated simply as "memory."

That translation is too narrow.

Chitta represents the accumulated mental field within which experience is retained, organized, and made available for future cognition.

It includes memory, but extends beyond memory alone.

Experiences leave impressions.

These impressions influence future perception, attention, reasoning, and behaviour.

Cognition is therefore never independent of previous experience.

Every act of thinking occurs against the background of an organized history.

Whether this idea corresponds directly to modern neuroscience is not the important question.

The more interesting question is architectural.

Should future AI systems possess a richer internal organization of memory than current models typically employ?

Perhaps memory should not merely retrieve information.

Perhaps it should continuously shape cognition itself.

Modern AI is already moving in this direction.

Retrieval-Augmented Generation (RAG), vector databases, long-context models, episodic memory modules, continual learning algorithms, and knowledge graphs all acknowledge an emerging reality.

Intelligence requires memory that actively participates in reasoning rather than passively storing information.

Long before digital computers, Indian philosophers recognized that memory is not simply a warehouse.

It is an active participant in cognition.

Ahamkara: The Self as a Functional Model

Perhaps the most controversial component of classical Indian cognition is **Ahamkara**.

The term is frequently translated as "ego."

Unfortunately, this translation often carries modern psychological associations that obscure its functional role.

Here, Ahamkara should not be understood primarily as pride or vanity.

Instead, it represents the organizing principle through which experience becomes centered upon an individual perspective.

Every intelligent system must distinguish, in some sense, between itself and its environment.

Autonomous robots must know which components belong to their own body.

Self-driving vehicles must distinguish their own motion from the movement of surrounding traffic.

Scientific reasoning requires distinguishing one's own hypotheses from external observations.

Even conversation requires maintaining continuity across multiple interactions.

Without some form of self-model, intelligent behaviour becomes remarkably difficult.

Modern AI increasingly confronts precisely this challenge.

Should future autonomous agents possess persistent identities?

Should they remember previous interactions?

Should they maintain long-term goals?

Should they distinguish between internally generated beliefs and externally verified facts?

These questions concern functional selfhood rather than consciousness.

Ahamkara offers an intriguing conceptual perspective.

It suggests that intelligent behaviour requires an organizing principle that integrates perception, memory, reasoning, and action around a coherent point of reference.

Whether this principle ultimately proves computationally useful remains an empirical question.

But the architectural insight is striking.

Selfhood may be viewed not merely as a philosophical mystery but as an important functional component of intelligent systems.

An Organized Society of Cognitive Functions

Taken individually, each of these four concepts is interesting.

Taken together, they become something much more significant.

They describe not isolated mental abilities but an interacting cognitive system.

Manas receives, filters, and coordinates incoming information.

Buddhi evaluates, discriminates, and decides.

Chitta provides the accumulated context of previous experience.

Ahamkara maintains continuity of identity and agency.

None operates independently.

Attention influences memory.

Memory guides reasoning.

Reasoning modifies future memories.

The self coordinates goals across time.

Every component both influences and depends upon the others.

This picture is surprisingly familiar.

Modern cognitive science increasingly views intelligence as an emergent property arising from interactions among multiple specialized systems.

Similarly, contemporary AI is moving toward architectures integrating perception, planning, retrieval, reasoning, external tools, and long-term memory.

Perhaps intelligence has always required organization more than uniformity.

Pull Quote

"Intelligence may emerge not from one powerful processor, but from many specialized processes working together."

Comparing Ancient Functional Concepts with Modern Cognitive Science

The comparisons proposed here should be interpreted as functional analogies rather than historical equivalences.

No claim is being made that ancient Indian philosophers anticipated cognitive neuroscience or artificial intelligence.

Instead, the comparison illustrates how different intellectual traditions sometimes identify similar functional problems.

Classical Concept	Functional Role	Modern Analogy
Manas	Coordination of sensory input and attention	Selective attention, attentional control, aspects of working memory
Buddhi	Judgement, discrimination, decision-making	Executive control, reasoning, cognitive flexibility

Chitta	Memory and accumulated experience	Long-term memory, episodic memory, continual learning
Ahamkara	Self-reference, agency, identity	Self-model, agent identity, self-representation

The purpose of this comparison is not to collapse one framework into the other.

Rather, it invites dialogue between conceptual traditions separated by centuries but united by a common interest in understanding cognition.

Toward an Indian Cognitive Architecture

Suppose an AI engineer were asked a deceptively simple question.

"What functional components should a generally intelligent system possess?"

Current AI research provides many answers.

Attention mechanisms.

Working memory.

Planning modules.

External tools.

Long-term memory.

Reasoning engines.

Self-models.

Goal management.

These components are usually discussed independently.

Classical Indian cognitive traditions suggest an alternative way of viewing the problem.

Perhaps they should not be considered isolated capabilities.

Perhaps they form an interacting architecture.

Imagine an intelligent system organized as follows:

Environment

↓

Perception



Manas

(Attention and Coordination)



Chitta

(Memory and Experience)



Buddhi

(Reasoning and Decision)



Ahamkara

(Self-model and Goal Maintenance)



Action



Feedback

This diagram is not proposed as a finished engineering design.

It is a conceptual hypothesis.

Its value lies in encouraging AI researchers to think architecturally rather than algorithmically.

The history of AI has often emphasized learning algorithms.

The coming decades may increasingly emphasize the organization of cognition itself.

Looking Ahead

Every scientific discipline occasionally benefits from ideas originating beyond its traditional boundaries.

Biology transformed computer science through evolutionary computation.

Neuroscience inspired artificial neural networks.

Physics shaped information theory.

Perhaps philosophy, too, can contribute—not by providing algorithms, but by suggesting new conceptual architectures.

Classical Indian philosophers were not designing intelligent machines.

Their objective was to understand the organization of human cognition.

Yet in pursuing that goal, they produced one of history's most sophisticated functional analyses of the mind.

Whether these ideas ultimately inspire future AI architectures remains to be discovered.

But they invite an important scientific question.

If intelligence is an organized interaction among specialized cognitive functions, then perhaps the future of AI depends not only upon building larger models, but upon building richer architectures.

That possibility leads naturally to the next article.

If cognition consists of interacting functional systems, how should those systems reason?

More than two millennia ago, the Nyāya philosophers developed one of the world's most sophisticated theories of logic, inference, and knowledge.

Remarkably, many of the questions they asked continue to challenge Artificial Intelligence today.

ICA Research Notebook

Functional Principle Introduced

Intelligence emerges through the interaction of specialized cognitive functions rather than a single undifferentiated processor.

Working Hypothesis

Future AI architectures that explicitly coordinate attention, memory, reasoning, and self-modeling may exhibit greater adaptability, interpretability, and long-term autonomy than architectures relying primarily on homogeneous computation.

Computational Prediction

Systems incorporating persistent memory, executive coordination, and explicit self-models will demonstrate improved continual learning, long-horizon planning, and context-sensitive reasoning.

Open Research Question

Can functional decompositions of cognition derived from classical philosophical traditions generate experimentally testable hypotheses for next-generation cognitive architectures?

Question for the Next Article

How should an intelligent system determine what is true?

The Nyāya school answered this question through a remarkably sophisticated theory of perception, inference, analogy, and testimony—ideas that continue to resonate with explainable AI, symbolic reasoning, and knowledge representation today.